Factors, Multiples & Primes

Year 4 · **Lesson Worksheet**

Worked Examples

Easy: List all the factors of 12.

Answer: 1, 2, 3, 4, 6, 12 (each pair multiplies to 12: 1×12, 2×6, 3×4).

Standard: List the first five multiples of 6.

Answer: 6, 12, 18, 24, 30 (keep adding 6).

Advanced: Find the common factors of 18 and 24.

Answer: Factors of 18: 1,2,3,6,9,18. Factors of 24: 1,2,3,4,6,8,12,24. Common factors: 1, 2, 3, 6.

Easy / Standard (15 questions)

1. Is 4 a prime number or not? Explain briefly.
2. List all the factors of 10.
3. Select ALL of these that are multiples of 8: A) 16 B) 24 C) 28 D) 32 E) 36
4. Is 33 a multiple of 5?
5. What is 7 squared?
6. Is 9 a prime number or not? Explain briefly.
7. Which of these is NOT a factor of 21? A) 1 B) 3 C) 5 D) 7
8. List the first five multiples of 7.
9. Is 34 a multiple of 10?
10. What is 5 squared? A) 10 B) 15 C) 20 D) 25
11. Is 10 a prime number or not? Explain briefly.
12. List all the factors of 8.
13. Which of these is the 5th multiple of 9? A) 36 B) 40 C) 45 D) 50
14. Is 31 a multiple of 4?
15. What is 4 squared? A) 8 B) 12 C) 16 D) 20

Standard (15 questions)

1. List all the prime numbers up to 25.
2. Is 9 a square number? What is √9? A) No; not applicable B) Yes; √9 = 2 C) Yes; √9 = 3 D) Yes; √9 = 4
3. Find the common factors of 12 and 18.
4. What is the lowest common multiple (smallest shared multiple) of 6 and 5?

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5. List all the factor pairs of 30.
6. Select ALL of these that are prime numbers: A) 41 B) 42 C) 43 D) 45 E) 51
7. Is 49 a square number? What is √49?
8. Find the common factors of 8 and 12.
9. What is the lowest common multiple (smallest shared multiple) of 2 and 4? A) 2 B) 4 C) 6 D) 8
10. List all the factor pairs of 36.
11. List all the prime numbers up to 20.
12. Is 81 a square number? What is √81? A) Yes; √81 = 7 B) Yes; √81 = 8 C) Yes; √81 = 9 D) No
13. Find the common factors of 20 and 30.
14. What is the lowest common multiple (smallest shared multiple) of 5 and 6? A) 11 B) 20 C) 25 D) 30
15. List all the factor pairs of 48.

Standard / Hard (15 questions)

1. Is 16 a prime number, a square number, or neither?
2. Two buses leave the station at the same time. One returns every 4 minutes, the other every 6 minutes. After how many minutes will they next both be at the station together? A) 4 B) 6 C) 10 D) 12
3. 3 and 5 are both prime numbers. What is 3 × 5, and is the answer odd or even?
4. 24 has how many factors in total? List them.
5. Find the highest common factor of 18 and 24. A) 3 B) 6 C) 9 D) 12
6. Is 25 a prime number, a square number, or neither?
7. 36 has 9 factors. Which of these are factors of 36? Select ALL that apply. A) 4 B) 5 C) 6 D) 9 E) 10
8. Two buses leave the station at the same time. One returns every 5 minutes, the other every 2 minutes. After how many minutes will they next both be at the station together?
9. 5 and 7 are both prime numbers. What is 5 × 7, and is the answer odd or even? A) 35, even B) 35, odd C) 12, odd D) 12, even
10. 30 has how many factors in total? List them.
11. Find the highest common factor of 15 and 20.
12. Is 27 a prime number, a square number, or neither? A) Prime B) Square number C) Neither D) Both prime and square
13. Two buses leave the station at the same time. One returns every 6 minutes, the other every 9 minutes. After how many minutes will they next both be at the station together?
14. 40 has 8 factors. Which of these are factors of 40? Select ALL that apply. A) 5 B) 6 C) 8 D) 10 E) 12
15. Find the highest common factor of 16 and 20.

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